

without spending ridiculous amounts of money – even with SpaceX lowering the costs drastically, it still costs over \$2,000 to send one kg to space. That's \$200,000, or a little over 1.05 cr to send a 100 kg human to space. But wait, that's just a naked human, if you want to send clothes, food, air for them to breathe, etc, then you will have additional weight. Multiply all the additional food weight by the number of days they need to spend in space, and the weight of the air, the air scrubbers and other life support systems, and you soon start to see why space is so expensive.

Nanotechnology will help in all of these areas, by improving food storage, creating better air and waste

displays, longer lasting batteries, thinner and lighter gadgets, etc., but all that is stuff we have come to take for granted. More cores per processors, higher transistor density and counts with lower power use... it's all happening already!

The biggest leap forward for computing will be quantum computing, which is something a true expertise in nanotechnology promises as an end stage goal. Many believe that we'd need a quantum computer before we can ever have true AI – suggesting that human consciousness has something to do with quantum effects. Thus, a quantum computing future might very well have robot intelligence, which would mean that we

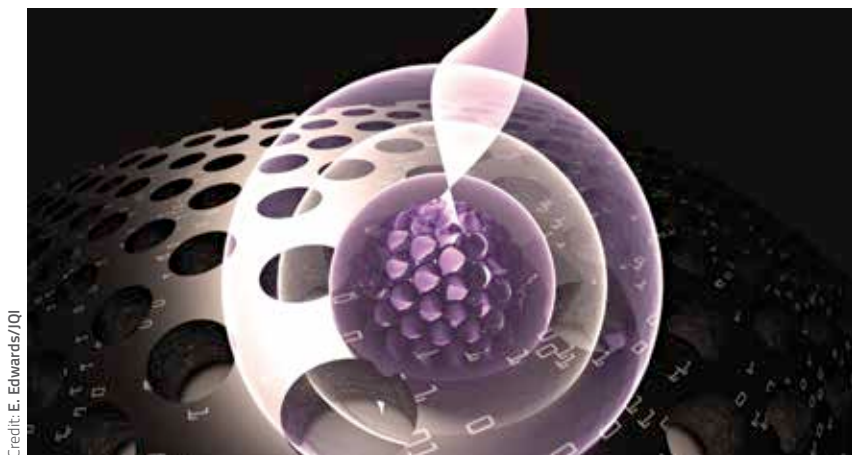
What about bombs that drop nanobots on top of a hapless population, all of them containing just the right dose of cyanide to kill you? And that's overt warfare, which is fast dying out anyway. The world has moved to covert warfare, and even cyber warfare, so what happens when we have little nanobots with the ability to dose you with anything without you even knowing about it?

The good news is that what nanotechnology can attack, it can also protect. Either way, the nano wars are coming, and we can be sure the military powers of the world (including India) already have technology that we do not know of, being used to protect us and our country.

Living with Nanotech

You can expect all interfaces to disappear, as more and more things become intuitive. Not only will the technology around us get smaller and more invisible, they will also be smarter thanks to being so integrated with us. If, as futurists predict, we all become part cyborg, as we interface with nanotechnology and other technologies, it will mean that technology will be more privy to our thoughts, behaviours and moods. Thus, when you come back home one night and are angry at something or someone, sensors will tell your house AI that you slammed the door, and your steps are slightly heavier as you stomp angrily around your home. It might choose to not remind you to pay your bills this evening with voice reminders, but might still go ahead and email you a reminder. That way you take out your anger on your cheap phone rather than your expensive house!

Whether we like it or not, life is set to become more interesting with more intrusive and pervasive technology than we ever thought possible! The people who already are overwhelmed and want to disconnect from technology will shudder at such a thought, but there are some of us who wish we could succumb to the grid right this minute, and inject some good old nano tech into our veins! Which type are you? 



Credit: E. Edwards/IQI

A single photon transistor made using a semiconductor chip. A photon stored in quantum memory is used to toggle the states of other photons.

scrubbers, meaning less weight to carry into space. Nanosensors will help control every aspect of a ship, and prevent small leaks, or warn about impending system failures. By making everything lighter and more durable (the calling card for everything nano) we will need to send much less structural weight of ships into space, which will mean either more important things can be sent up, or else ships will be able to be bigger, and thus store more essentials and thus, travel further for longer.

Computing

Honestly, not much will change for computing, because we're already living in the nano world when it comes to technology. Yes we will get better screens, higher resolution

could send off robots to truly explore the solar system, which many believe is the only way humanity is ever going to explore the cosmos – living vicariously through our creations...

Warfare

This is where everything goes off the rails and gets really, really scary. It's not even hard to do, really, because all you have to do is look at every single positive thing we have mentioned through the course of this article and consider that to be something that your worst enemy controls. Got nanobots running through your veins trying to attack plaque? What happens if someone who wants you dead gains control of those nanobots? Imagine a quantum computing AI that is programmed to wipe out humanity?